American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
4	1/28/26	9 am	20 min	⦿ Advisor ○ Students	Lynn Ariss

Attendees:

Raghad Al Agha, Lynn Ariss, Hadi Elham, Claritta Khoury

Briefly summarize the main discussions during the meeting:

The meeting reviewed key technical developments across sensing, hardware control, software integration, and data security. The team confirmed that the BPM-based processing for systolic and diastolic signals is operating correctly and discussed battery regulation strategies, ensuring the Arduino can safely manage motor activation while monitoring voltage levels to prevent unsafe discharge. They addressed resistance inconsistencies within the Velostat pressure matrix and validated successful live data streaming to the application, including real-time 30-second visualization and encrypted transmission. The heatmap implementation was examined using a row-column scanning approach (digital columns and analog rows), with Python-based visualization tested for analog data. The team also considered expanding matrix resolution through multiplexers while maintaining compact hardware integration. Finally, encryption strategies and secure per-user key management were discussed to enhance system security.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team concluded that BPM signal processing, real-time visualization, and encrypted communication are functioning reliably, and that the heatmap approach using scanning logic and multiplexers is technically viable. Further optimization of the pressure matrix layout and hardware integration will continue to improve compactness and performance. Battery regulation and motor control mechanisms are currently acceptable but require additional validation under real-use conditions. It was agreed that encryption keys should not be hardcoded and must instead be securely generated and managed per user within the application. Overall, the project is steadily advancing toward a fully integrated, secure, and scalable wearable system, with the next phase focused on hardware stabilization and refining the multiplexed sensing architecture.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Build and test the charging circuit, try finalizing the flex circuit testing accurately, work on a higher resolution heatmap generation, work on detailed circuit schematics for the circuits being used so far	Raghad Al Agha	Feb 18
Continue hyperparameter tuning and performance evaluation, implement secure per-user AES key generation derived from a protected master key without embedding keys in the application code, and further develop and test the CSV data export functionality.	Lynn Ariss	Feb 18
Continue Velostat heat-map development, explore design alternatives, build charging circuit, and configure flex sensors.	Hadi Elham	Feb 18
Develop the algorithm for detecting peaks in the PPG sensor data	Claritta Khoury	Feb 18